

Robotics in India: What's Cooking?



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Today aerial, land and aquatic forms of robots are used for diverse applications in medical, military, industrial, space, entertainment and domestic market sectors. All these robots might look very different from each other but internally these all follow the same key paradigm that defines a robot—Sense, Think and Act. Robots must be able to sense the world, process the information and perform necessary actions. “The Sense, Think and Act paradigm is followed by robots in any situation, be it cleaning rivers, bomb disposal or flying. The only difference is in the implementation of the paradigm,” says Aakash Sinha, CEO and founder, Omnipresent Robot Technologies Pvt Ltd.

In this article, we highlight some of the recently-launched smartmachines across the world that are implementing the Sense, Think and Act paradigm in different ways to automate tasks, while focusing on the ecosystem for robotics in India.

Components on the rise for robotics applications

The sense, think and act tasks performed by a robot need their own set of components. Below are some components that are increasingly being embedded in intelligent machines today.

To sense. A lot of sensors are available off-the-shelf today. For example, SRS series ultrasonic sensors are very popular for building small robots, informs Sinha. Radar and lidar sensors are also widely used for various applications. Sinha adds, “Lidar sensors made by SICK Group or Velodyne are widely used, the latter being expensive but very good for applications like terrain mapping.”

To think. There is a plethora of single-board computers as well as auto pilots available in the market. Pixhawk auto pilot is popularly used in a lot of robotic vehicles, including the ones that operate on land, water and air. Similarly, there are many single-board computers readily available.

For building basic robots, one can start off with Raspberry Pi 2, Arduino or BeagleBone boards, informs Sinha. He says, “We have used Radxa Rock for some of our designs. It is quite powerful as it comes equipped with a quad-core processor, 2GHz of random access memory (RAM) and NAND flash built into it, and is very good for advanced robots.”

To act. Servomotors are a key element in robotics and the industrial automation space. Motors are made as light in weight as possible but meet the heavyweight expectations from the industry today. There are some very promising introductions in

Automated lockers powered by robotic arm

Traditional lockers in banks need to be manually opened to access and retrieve valuables and other particulars stored in it. There are two keys for such lockers, and in a typical situation, a bank representative inserts the key in one slot and allows the customer to access the locker with the other key that is in his or her possession. Accessing lockers in total privacy without the intervention of any bank official could soon be possible in banks across the country.



An artistic representation of Smart Vault facility provided by ICICI Bank

ICICI bank recently introduced Smart Vault, a futuristic locker facility in New Delhi. It is powered by robotics and automation, and facilitates customers to access the locker any time as per their convenience and in a completely private lounge area. With a robotic arm placed in the bank's vault, it fetches the locker and presents it to the customer after he or she passes a multi-level security check that demands for biometric identification and card details. The robotic arm uses the customer's co-ordinates and RFID technology to identify the right locker.